

Phantom Menehune Zookeeper Robot

Cost Analysis per Unit — CSMFAB MECH003

Category: Robotics — Protective Swarm **Unit Basis:** 1 robot (complete 90cm unit) **Date:** June 2026 | **Document Type:** CSMFAB Cost Study V2.0

1. Product Summary

This cost analysis is derived from the V2.0 specification and v1to2 versioning document for this product. All material costs reflect 2026 market pricing. Three production scales are costed: - **Prototype:** 1-5 units (single-setup, high labor, no tooling amortization) - **Small Batch:** 100-500 units/year (semi-tooled, moderate labor) - **Production Volume:** 5,000-50,000 units/year (fully tooled, optimized labor)

2. Bill of Materials (BOM)

Component	Quantity/Unit	Unit Price	Extended Cost
BFRP chassis (Elium®, 90cm scale)	12 kg	\$25.00/kg	\$300.00
KNbO3-BaTiO3 actuators (12 joints)	1.5 kg	\$180.00/kg	\$270.00
PVDF-TrFE footstep harvesting stack	0.2 kg	\$350.00/kg	\$70.00
Si3N4 gear and bearing set	1 kg	\$90.00/kg	\$90.00
LiFePO4 power storage (200Wh)	1.6 kg	\$100.00/kg	\$160.00
LoRa mesh radio (256-node capable)	0.05 kg	\$25.00/kg	\$1.25
LiDAR sensor (PEEK-housed)	0.3 kg	\$350.00/kg	\$105.00
YInMn Blue chassis glaze	0.5 kg	\$75.00/kg	\$37.50
MRAM offline operation storage	0.01 kg	\$80.00/kg	\$0.80
Soft-robotic pressure actuator kit (grippers)	0.8 kg	\$280.00/kg	\$224.00
BOM Subtotal			\$1,258.55

3. Labor, Process & Overhead Costs

Cost Element	Prototype	Small Batch	Production Volume
BOM Materials	\$1,258.55	\$1,032.01	\$818.06
Direct Labor	\$6,800.00	\$3,800.00	\$2,200.00
Overhead (15% of labor)	\$1,020.00	\$570.00	\$330.00
Quality Control & Testing	\$816.00	\$304.00	\$110.00

Cost Element	Prototype	Small Batch	Production Volume
Packaging & Logistics	\$37.76	\$31.46	\$25.17

Tooling & Equipment Notes: BFRP mold: \$28,000 NRE. Much simpler actuator set than Phantom MK-1.

4. Per-Unit Cost Summary

Scale	Total COGS	Suggested MSRP Range	Gross Margin
Prototype (1-5 units)	\$32,000	\$51,200 – \$70,400	35-55%
Small Batch (100-500/yr)	\$22,000	\$33,000 – \$44,000	33-50%
Production Volume (5K-50K/yr)	\$15,000	\$21,000 – \$27,000	28-45%

Market Reference: \$18,000-28,000 per unit (wildlife conservation NGO + military non-lethal patrol market)

5. Key Cost Drivers (V2.0)

Driver	Impact	Mitigation
ZrB2/Si3N4 UHTC ceramics	HIGH — specialty powder \$350+/kg	Bulk procurement; domestic synthesis pilot (Japan NIMS CO2 route)
MXene Ti3C2Tx	VERY HIGH — \$2,000/kg at 2026 scale	Tile pattern minimizes usage; price projected -60% by 2028 as production scales
Elium® thermoplastic BFRP	MEDIUM — recyclability adds value; \$12/kg resin	Phoenix Protocol recycling recovers 100% monomer; net cost offset
Tooling NRE	HIGH for first production run	Paid by pre-orders or grant funding; amortized quickly
Flash Sintering process	MEDIUM — shares SPS equipment	Regional ceramic sintering facility sharing model

6. Phoenix Protocol Circular Economy Value

At end-of-life, material recovery adds back-value: - **Recovered material value per unit:** \$226 – \$440 - **Recovery reduces net lifecycle cost by:** 15-30% - **Critical minerals (In, Y, Co, Ti) recovered** via ionic liquid extraction